

Latent World Modeling for Deformable Object Manipulation

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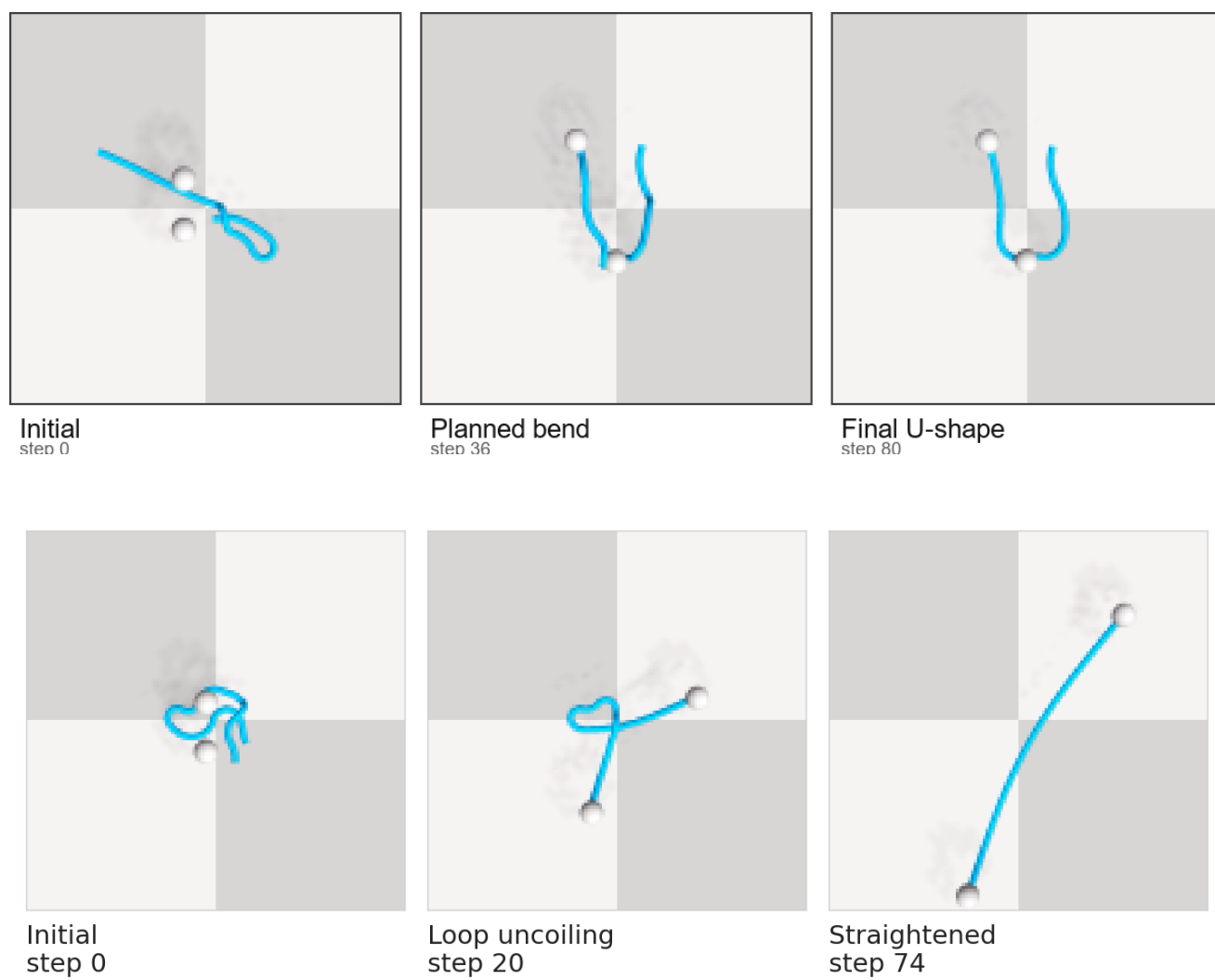


Introduction

Deformable objects such as ropes, cloth, cables, and tissue are essential in robotics but remain difficult to manipulate because their state is high-dimensional, partially occluded, and governed by contact-rich dynamics. Classical simulator-based planning can be accurate but is too slow for real-time control, while recent vision-language-action policies often rely on billion-parameter backbones.

Our question: Can a compact *latent* world model trained from pixels support practical planning for deformable manipulation?

SoftGym Rope Tasks



We evaluate goal-conditioned rope manipulation from 128×128 top-down RGB, two-picker actions $a_t = [v_1, g_1, v_2, g_2]$, and proprioception. RopeFlatten tests straightening by endpoint distance; U-shape tests whether latent MPC can follow a more constrained global target.

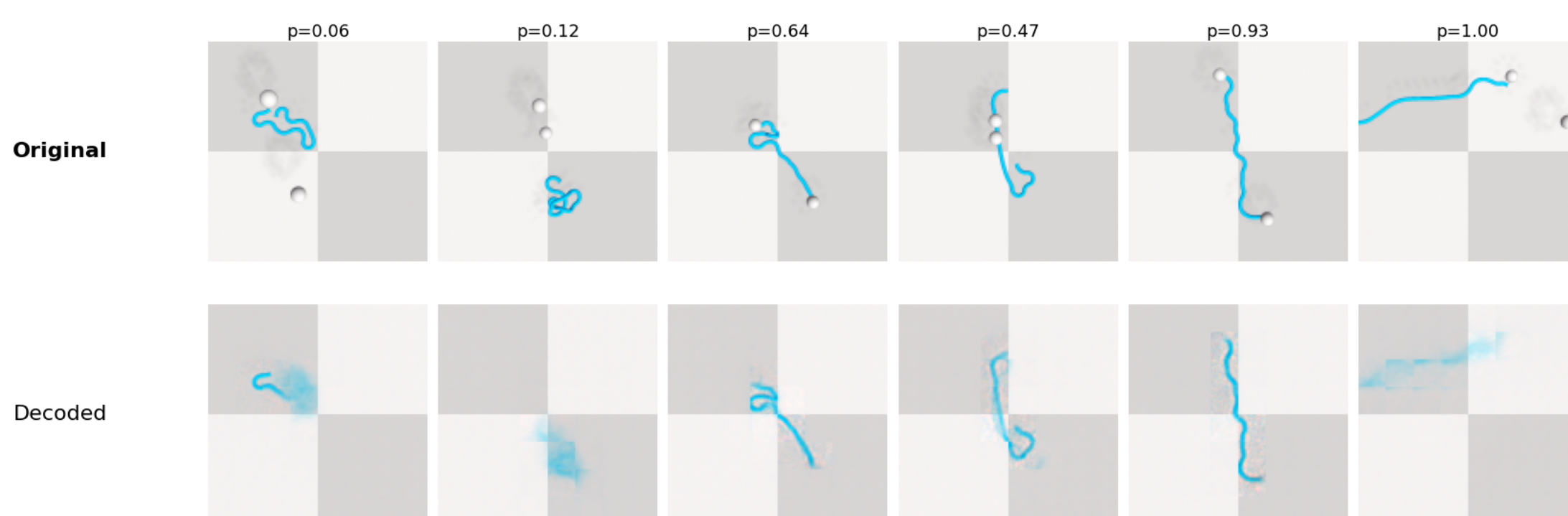
Dataset & Approach

We collect 19,031 offline episodes (1.45M frames) from a near-optimal CEM oracle plus three lower-cost policies that broaden state-action coverage for planning.

Source	Episodes	Perf.
CEM expert oracle	4,031	0.994
Geometric solver	5,000	0.905
Manipulative heuristic	5,000	0.448
Random exploration	5,000	0.409
Full mixed set	19,031	0.674

Mixed data exposes the model to missed grasps, collisions, and folded rope states that expert-only demonstrations rarely visit. Our 18.6M-parameter LeWM-style JEPA predicts future latents, and CEM-MPC plans in the 192-D latent space using distance to a goal image.

Latent Visualization



A diagnostic decoder, used only for visualization, shows that the 192-D latent preserves rope extent, curvature, and endpoint layout.

Latent World Model

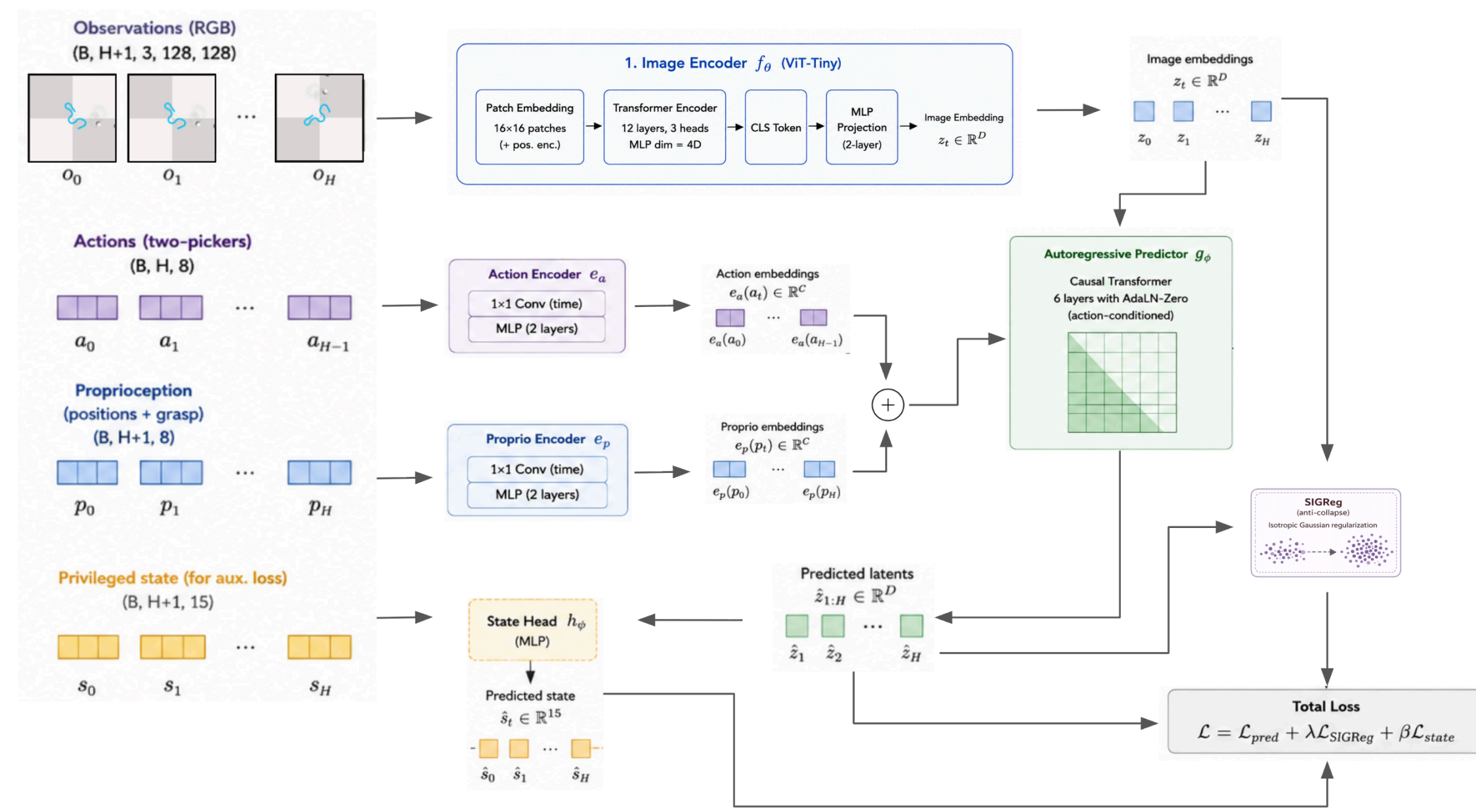


Image encoder f_θ

A ViT-Tiny encoder splits each 128×128 frame into 16×16 patches and produces a 192-D [CLS] latent.

Action-conditioned predictor g_ϕ

A 6-layer causal transformer with AdaLN-Zero conditioning predicts $\hat{z}_{t+1}$ from $H=3$ latent history, actions, and proprioception. Proprioception conditions the dynamics but stays out of z_t , so latent goal matching focuses on rope geometry rather than gripper pose.

Auxiliary state head h_ψ

During training, a small MLP predicts a 15-D privileged state from z_t to ground the latent in rope center of mass, bounding box, and picker state. This head is discarded at inference.

Training Objective

Next-latent prediction is combined with anti-collapse regularization and auxiliary geometric supervision:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{pred}}}_{\text{next-latent MSE}} + \lambda \underbrace{\mathcal{L}_{\text{SIGReg}}}_{\text{anti-collapse}} + \beta \underbrace{\mathcal{L}_{\text{state}}}_{\text{geometry head}}.$$

Following LeJEPa, **SIGReg** projects each batch onto $M=1024$ random one-dimensional directions and matches those projections to $\mathcal{N}(0, 1)$. This prevents collapse and keeps latent distances well-scaled for MPC. We train with $\lambda=0.09$, $\beta=0.5$, AdamW, learning rate 5×10^{-5} , batch size 128, bf16 precision, and 100k steps on one H100.

Latent-Space Planning (CEM-MPC)

A goal image is encoded once as z_g . At each replan step, CEM samples 300 action sequences with horizon $K=5$, rolls out the predictor autoregressively, and scores each plan by terminal latent distance:

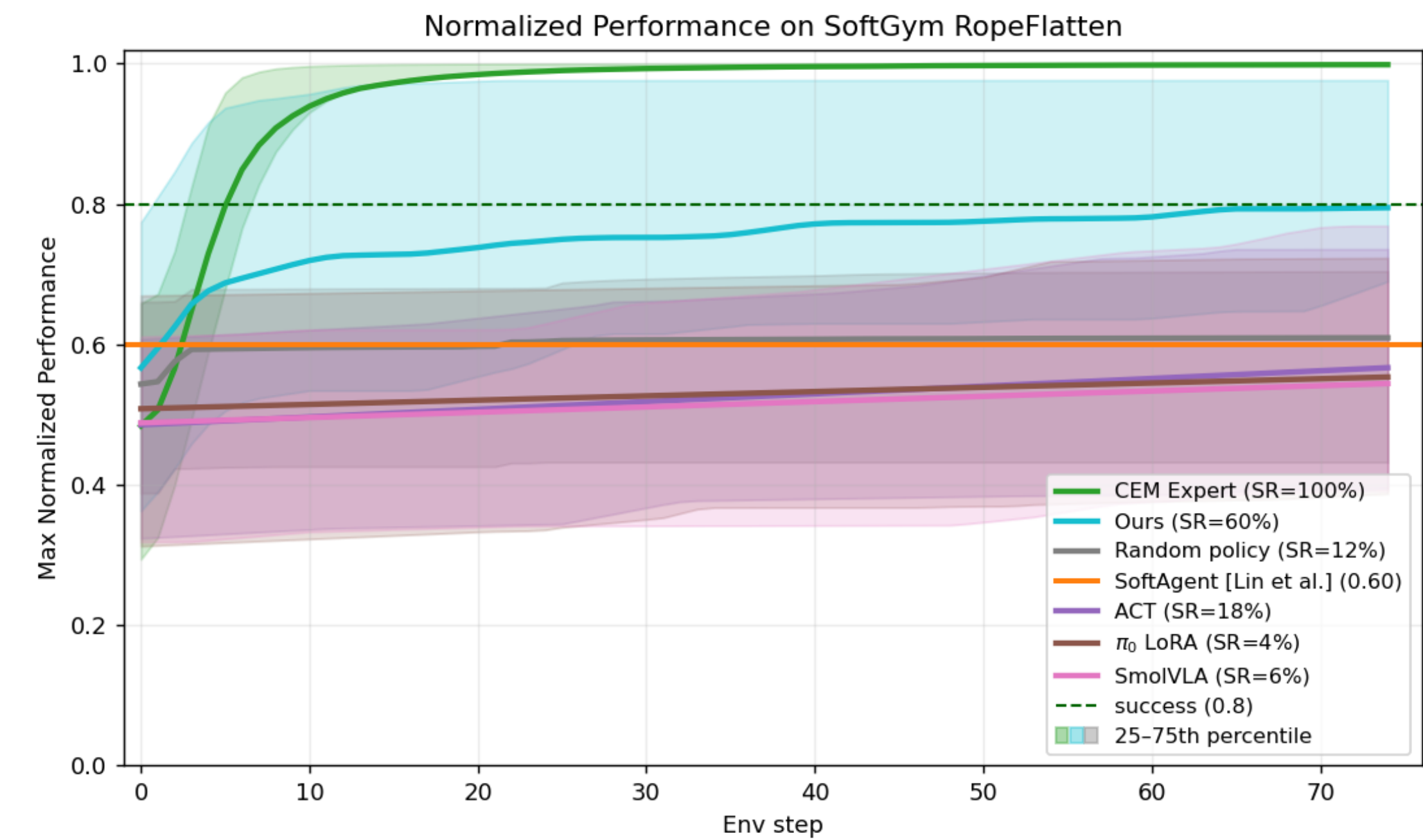
$$J(A_{0:K-1}) = \|\hat{z}_{t+K} - z_g\|_2^2.$$

Success metric

Normalized RopeFlatten performance is computed from endpoint distance; a rollout succeeds if final performance exceeds 0.8.

The distribution is refit to the top-30 plans for 30 iterations, then the first action is executed before replanning from the real observation. Each real action evaluates 9,000 candidate plans, corresponding to 45,000 imagined predictor steps, entirely in 192-D latent space with no pixel reconstruction or simulator reward.

Headline Result



Policy	Params	Perf.	Latency
LeWM-MPC	18.6M	0.848	822 ms
ACT	52M	0.542	26 ms
SmoVLA	450M	0.444	283 ms
π_0 LoRA	3B	0.432	309 ms

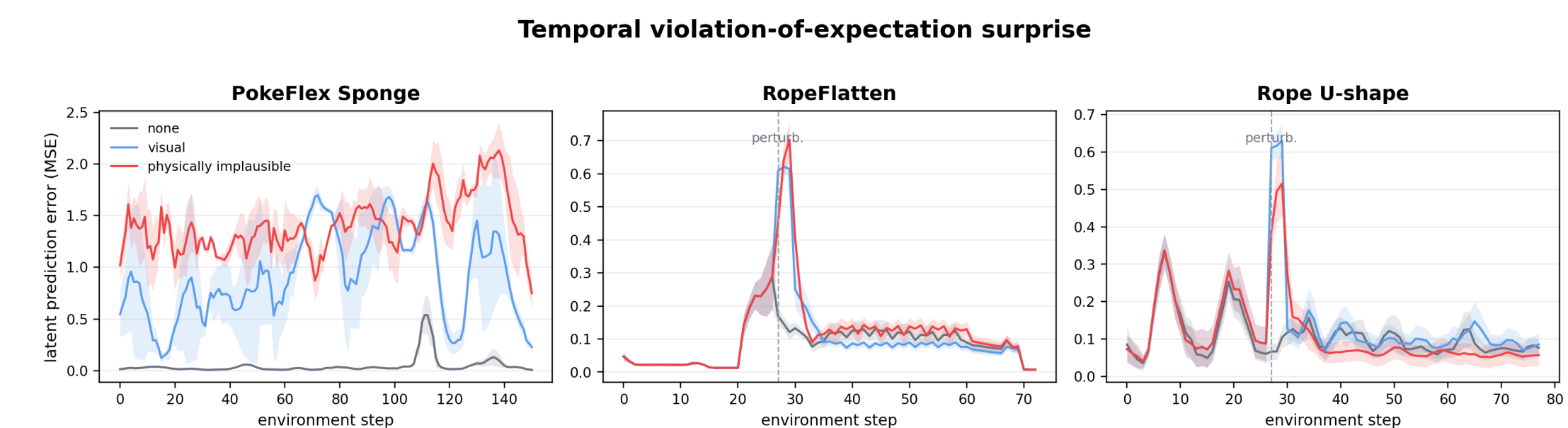
LeWM-MPC reaches **60% success**, outperforming ACT and pretrained VLAs up to 150× larger.

Ablation: Prediction MSE vs. Task Success

Run	Pred MSE ↓	SR % ↑
mix19k_aux	0.0152	60
mix19k_base (–aux)	0.0168	46
CEM-only (–aux)	0.0188	38
Red picker, –proprio	0.0415	12

The auxiliary geometry head improves success from **46%** to **60%** with only a small MSE change, so better planning comes from more useful latents rather than lower prediction error alone.

Violation-of-Expectation Test



Visual and physical perturbations produce surprise spikes near the perturbation step, then fall as the perturbed observation enters the context. This suggests the model learns expected deformable dynamics, not only static appearance.

Conclusions & Future Work

Compact JEPA-style world models extend from rigid-body control to contact-rich deformable manipulation from pixels. At 18.6M parameters, LeWM-MPC outperforms image-based RL baselines and direct policies/VLAs up to 150× larger. Future work will compare against V-JEPA 2-AC, improve long-horizon U-shape maintenance, and transfer to real-hardware cloth folding.